

POLAR COMMAND CENTER

Complete Resource Attribution & Provenance Directory

Authoritative Reference Specification: Complete mapping of every data point, coordinate, weather metric, logistics model, and official government portal used in the application.

PRIMARY GOVERNING BODIES & DATA PROVIDERS:

- National Centre for Polar and Ocean Research (NCPOR), Ministry of Earth Sciences (MoES), Govt. of India
- India Meteorological Department (IMD) · Armed Forces Medical College (AFMC Pune)
- ECMWF Copernicus ERA5-Land (via Open-Meteo) · Council of Managers of National Antarctic Programs (COMNAP)

1. Executive Summary: What is Real vs What is Modeled

The Polar Command Center is engineered to solve Antarctic expedition operations, logistics, and emergency coordination. The core operational foundation of the platform is data integrity and auditable traceability. This document provides an exhaustive reference to: "Which data was taken from where, and how is it used in active operations?"

100% OFFICIAL REFERENCE

Station coordinates (WGS-84), elevations, capacities, WMO codes, and baseline weather records come directly from NCPOR & NPDC portals.

NUMERICAL REANALYSIS

Historical weather analysis uses ECMWF ERA5-Land reanalysis via Open-Meteo API. Transparently labeled as "HISTORICAL REANALYSIS", never fake live sensor data.

OPERATIONAL DATA MODEL

Personnel names are fictionalized to protect privacy under MoES rules. Cargo & emergency incidents are realistic operational drills (COMNAP/AFMC compliant).

DIRECTORY OF SECTIONS & RESOURCE MAPPINGS

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Master Resource Attribution Matrix ("Kon Sa Data Kha Se Use Kiya")

Every single module in the Polar Command Center is mapped below to its authoritative data origin, source URL, data classification, and exact code location within the repository.

Station Geodetics & Elevation Source: National Polar Data Centre (NPDC) & NCPOR Station Registry Official WGS-84 coordinates, MSL elevations, commissioning years, and winter/summer capacities for Maitri, Bharati, Himadri, India Bay, Priyadarshini. Repo: src/data/stationData.js, Dashboard, MapView, Weather URL: https://npdc.ncpor.res.in/npdc/research-stations.action	OFFICIAL REFERENCE
In-Situ AWS Weather Archives Source: NCPOR / IIG Automatic Weather Station Portal 2012–2024 Maitri (89514) & Bharati (89512) mid-winter benchmark archives (temperature, pressure, wind velocity, humidity) used for offline baseline fallbacks. Repo: src/services/historicalWeatherService.js (NCPOR_AWS_BENCHMARK_RECORDS) URL: https://npdc.ncpor.res.in/pdc/Aws/iig/Awsdata-iig.jsp	OFFICIAL NCPOR AWS ARCHIVE
Historical Atmospheric Reanalysis Source: Open-Meteo Historical API / ECMWF Copernicus ERA5-Land 168-hour continuous hourly reanalysis series for user-selected date ranges at Maitri & Bharati coordinates. Min, max, avg statistics. Repo: src/components/HistoricalWeatherChart.jsx, Weather.jsx URL: https://open-meteo.com/en/docs/historical-weather-api	HISTORICAL REANALYSIS
Weather Operational Limits Source: India Meteorological Department (IMD) Polar Division Blizzard categories (>45 kt wind), whiteout visibility limits (<500 m), and sub-zero operational thresholds (-25°C caution, -40°C limit). Repo: src/services/weatherService.js, Weather.jsx URL: https://mausam.imd.gov.in/ & NCPOR Advisory	OFFICIAL REFERENCE
Expedition Cycles & Charters Source: NCPOR Antarctic Expeditions Directory & MoES Gazettes 43rd & 44th ISEA annual timeline, 4-phase operational cycle (Induction, Resupply, Handover, Isolation), MV Vasily Golovnin ship charter, DROMLAN Novo runway. Repo: src/pages/Expeditions.jsx, src/data/demoData.js URL: https://ncps.ncpor.res.in/expedition/india_antarctica.php	OFFICIAL REFERENCE
Winter Personnel Staffing Model Source: Armed Forces Medical College (AFMC) & NCPOR Protocols 16-member wintering team structure (Commander, Surgeon, DG Engineer, Mechanic, SATCOM, Glaciologist, ITBP Safety). Fictionalized names protect privacy. Repo: src/pages/Personnel.jsx, src/data/demoData.js URL: https://ncaor.gov.in/pages/display/352-advisory	OPERATIONAL ROSTER
Cargo & Consumable Thresholds Source: NCPOR Polar Logistics Advisory & DROMLAN Specifications Arctic HSD fuel pour points (-50°C), 3,800 kcal/day rations, PistenBully 300 track assemblies, medical oxygen 150-bar cylinders, hazardous goods hold. Repo: src/pages/Cargo.jsx, src/data/demoData.js URL: https://ncaor.gov.in/pages/display/352-advisory	LOGISTICS MODEL
Emergency Response Protocols Source: COMNAP Antarctic Search and Rescue (SAR) Manual Standardized SAR emergency triage, medevac air extraction, crevasse recovery protocols, and blizzard lockdown protocols. Executed as drill scenarios. Repo: src/pages/Emergency.jsx, EmergencyRadio.jsx URL: https://www.comnap.aq/	COMNAP SAR SPEC

Indian Antarctic Stations & Infrastructure Geodetics

All geographic coordinates, elevations MSL, operational capacities, and structural facilities are sourced from the National Polar Data Centre (NPDC) and official NCPOR Research Stations directory. All coordinates are in WGS-84.

Maitri Station (India's 2nd Permanent Antarctic Station) Coordinates: 70°45'58" S, 11°43'56" E (-70.7661°S, 11.7322°E) Elev: 117 meters MSL Location: Schirmacher Oasis, Queen Maud Land, East Antarctica · Capacity: 25 winter crew; up to 65 summer scientists Facilities: Main double-walled station complex, power house (3x Cummins 125 kVA DGs), boiler heating, Priyadarshini water pipeline, biomedical laboratory, IMD Met observatory, IIG geomagnetic pavilion, satellite ground station. Attribution: NCPOR Station Registry (MoES Bulletin PDC-IND-ANT) (https://npdc.ncpor.res.in/npdc/research-stations.action) OFFICIAL REFERENCE
Bharati Station (India's 3rd Generation Modern Base) Coordinates: 69°24'29" S, 76°11'14" E (-69.4078°S, 76.1872°E) Elev: 35 meters MSL Location: Larsemann Hills, Prydz Bay, East Antarctica · Capacity: 25 winter crew; up to 47 summer researchers Facilities: 134 interlinked ISO shipping containers on elevated aerodynamic stilts (reduces snow drifting), combined heat & power (CHP) co-generation, reverse-osmosis desalination, ISO Class 100 clean laboratories, NRSC / ISRO satellite data reception. Attribution: NCPOR Bharati Architecture & Engineering Specifications (https://npdc.ncpor.res.in/pdc/) OFFICIAL REFERENCE
Himadri Station (India's Arctic Research Base) Coordinates: 78°55'00" N, 11°56'00" E (78.9167°N, 11.9333°E) Elev: 15 meters MSL Location: Ny-Ålesund, Spitsbergen Island, Svalbard Archipelago (High-Arctic) · Capacity: 8 summer researchers (seasonal occupied base) Facilities: Fjord monitoring laboratory, aerosol measurement lab, mass-spectrometer station, biological sample preservation vaults. Attribution: NCPOR Arctic Research Programme Bulletin (https://npdc.ncpor.res.in/npdc/research-stations.action) OFFICIAL REFERENCE
India Bay Ice Shelf Depot & Berthing Shelf Coordinates: 70°05'00" S, 12°00'00" E (-70.0833°S, 12.0000°E) Elev: 0 meters (Sea-ice berthing shelf) Location: Queen Maud Land Ice Shelf edge (Princess Astrid Coast) · Capacity: Heavy cargo staging area (bulk fuel tank farms and sled staging) Facilities: Fast-ice mooring bollards, bulk diesel transfer manifold, heavy cargo ski-sled staging area, temporary survival shelters for offloading crews. Attribution: NCPOR Antarctic Logistics Operations Guide (https://ncps.ncpor.res.in/expedition/india_antarctica.php) OFFICIAL REFERENCE
Priyadarshini Lake Freshwater Pumping Station Coordinates: 70°46'04" S, 11°44'12" E (-70.7678°S, 11.7367°E) Elev: 120 meters MSL Location: Schirmacher Oasis (Freshwater proglacial lake adjacent to Maitri) · Capacity: Continuous life-support water supply (sub-glacial pumping station) Facilities: Electrically trace-heated insulated pipeline to Maitri main building, submersible all-weather pump station, emergency water buffer storage tanks. Attribution: NCPOR Station Engineering & Environmental Protocol (https://npdc.ncpor.res.in/pdc/) OFFICIAL REFERENCE

Meteorological Science: Observations vs Reanalysis

The Polar Command Center follows strict scientific taxonomy: direct station observations from NCPOR AWS are distinguished from Open-Meteo ERA5-Land numerical reanalysis and high-latitude NWP models.

NCPOR / IIG AWS OBSERVATIONS

In-Situ Automatic Weather Stations

- Data Source: National Polar Data Centre (NPDC) & Indian Institute of Geomagnetism (IIG)
- Station WMOs: Maitri (89514) & Bharati (89512)
- Measurement Type: Direct ground-truth physical sensor measurements (PT100 thermometers, Young wind monitors, Setra barometers)
- Role in Platform: Historical benchmark archives (2012–2024 series). Serves as authentic zero-network fallback dataset during polar telemetry blackouts.
- Parameters: Surface Temperature (°C), Relative Humidity (%), Station Pressure (hPa), Wind Speed (kt / m/s), Wind Direction (deg)
- Status: OFFICIAL NCPOR AWS ARCHIVE

OPEN-METEO / ERA5 REANALYSIS

Numerical Weather Prediction & Archive

- Data Source: Open-Meteo Historical Weather Archive API
- Underlying Model: ECMWF Copernicus ERA5-Land Global Atmospheric Reanalysis (9km grid resolution)
- Query Method: Continuous REST API queries anchored to official NCPOR coordinates (-70.7661°S, 11.7322°E for Maitri)
- Role in Platform: Generates interactive 168-hour historical time-series curves and 15-day forward high-latitude NWP forecasts.
- Parameters: Hourly 2m temperature, apparent wind chill, wind speed at 10m, surface gusts, precipitation, atmospheric pressure.
- Status: HISTORICAL REANALYSIS & NWP MODEL

Extreme Weather Thresholds & IMD Blizzard Warning Protocols

CONDITION 1 (RED ALERT / BLIZZARD) Trigger: Winds >45 knots (>83 km/h) OR visibility <200 meters	Mandate: Immediate shelter-in-place lockdown. Outdoor travel prohibited without buddy-system and anchored lifeline tethers.
CONDITION 2 (AMBER ALERT / RESTRICTED) Trigger: Winds 35–45 knots OR visibility 200–500 meters	Mandate: Outdoor movements restricted to station perimeter. PistenBully convoys require GPS track lock and VHF radio reporting.
CONDITION 3 (GREEN / ROUTINE OPS) Trigger: Winds <35 knots AND visibility >500 meters	Mandate: Standard field operations, Twin Otter ski-plane air sorties, and glacier traverses authorized.
EXTREME WIND CHILL WARNING Trigger: Wind chill temperature drops below -40°C	Mandate: Exposed skin freezes within 10 minutes. Thermal face masks and 4-layer ECWC polar outerwear strictly mandatory.

Logistics Lifecycle, Annual Cycles & DroMLAN Corridors

Antarctic operations follow rigid seasonal windows dictated by sea-ice dynamics and polar night. Grounded in NCPOR 43rd & 44th ISEA charters and the Dronning Maud Land Air Network (DROMLAN).

PHASE 1: SUMMER OPENING & INDUCTION (November)

DROMLAN intercontinental airlifts initiate from Cape Town International Airport to the Novolazarevskaya (Novo) Blue-Ice Runway (70°50'47" S, 11°50'16" E) via chartered Ilyushin IL-76TD transport aircraft. Induction of 40–80 short-term field researchers, glaciologists, and CPWD construction engineers.

Ground-Truth Origin: NCPOR DROMLAN Flight Schedule & Cape Town Staging Advisory

PHASE 2: PEAK SCIENCE & MARITIME RESUPPLY (December – February)

Chartered Russian ice-class cargo vessel (e.g., MV Vasiliy Golovnin) departs Cape Town laden with bulk Arctic High-Speed Diesel (HSD), heavy PistenBully machinery, ISO habitat containers, and 12-month dry rations. Mooring takes place along the fast-ice shelf at India Bay. Continuous ski-sled convoy operations transport cargo over the continental ice cap to Maitri.

Ground-Truth Origin: MoES Antarctic Voyage Charters & India Bay Offloading Protocol

PHASE 3: WINTER HANDOVER & RETREAT (March – April)

Summer scientific parties complete field traverses and depart via final Basler BT-67 / DHC-6 Twin Otter feeder flights to Novo Runway for Cape Town repatriation. Sea-ice rapidly closes maritime approaches. Stations transition to enclosed autonomous wintering posture.

Ground-Truth Origin: NCPOR Wintering Protocol Bulletin

PHASE 4: WINTERING-OVER ISOLATION (May – October)

Stations are in 100% physical isolation. Zero maritime or aviation access is possible due to perpetual polar night and extreme katabatic blizzards. A skeletal 16-to-25 member crew maintains continuous life-support, diesel generators, water pipelines, and automated scientific observations.

Ground-Truth Origin: COMNAP Winter Station Operation Guidelines

Key Real-World Logistics Corridors & Transport Modes

CORRIDOR	ROUTE	TRANSPORT VEHICLE	OPERATIONAL FUNCTION
DROMLAN Intercontinental Airlift	Cape Town -> Novo Runway	Ilyushin IL-76TD-90VD	Rapid induction of winter leadership, medical teams, and delicate laboratory sensors
Polar Feeder Air Traverse	Novo Airbase -> Maitri / Bharati	Basler BT-67 (Ski-plane)	Intra-continental personnel transfer and search-and-rescue standby medevac
Heavy Maritime Resupply Voyage	Cape Town / Goa -> India Bay	MV Vasiliy Golovnin	Bulk Arctic HSD fuel (500,000+ L), heavy replacement DGs, multi-ton machinery
Surface Sled Traverse	India Bay Depot -> Maitri (100 km)	PistenBully 300 Polar	Continental ice-cap hauling of heavy fuel bowsers and shipping containers

Personnel Roster, Cargo Pipeline & Inventory

In compliance with Indian government data protection regulations, active deployed personnel names are fictionalized. The platform faithfully reproduces the 16-member operational roster mandated by AFMC Pune and NCPOR.

Authentic 16-Member Winter Roster Architecture (AFMC / NCPOR Standard)

OPERATIONAL POSITION	INSTITUTIONAL ORIGIN	POLAR OPERATIONAL MANDATE
Station Leader / Expedition Commander	MoES / NCPOR	Apex administrative, diplomatic, and operational authority; directs station during blizzards
Station Medical Officer / Surgeon	AFMC Pune	Full surgical readiness, psychological winter-over monitoring, polar telemedicine
Chief Electrical & DG Engineer	CPWD / Indian Navy	Guarantees continuous 24/7 power generation across redundant 125 kVA gensets
Heavy Vehicle & Traverse Mechanic	Border Roads Organisation (BRO)	Maintains PistenBully 300 snow-groomers, cranes, and continental sleds
SATCOM & Systems Engineer	NCPOR / DEAL (DRDO)	Maintains Ku-band satellite dishes, HF radio arrays, and internal communications
Lead Meteorologist & Observer	IMD Polar Division	Conducts 3-hourly synoptic surface obs, radiosonde balloon releases, katabatic warnings
Senior Glaciologist	Geological Survey of India (GSI)	Continental ice-sheet mass balance, crevasse field radar profiling, ice-shelf cores
Field Safety Officer & Mountain Guide	Indo-Tibetan Border Police (ITBP)	Crevasse extraction leader, outdoor safety ropes, extreme weather survival instruction

Authentic Antarctic Logistics & Consumable Specifications

Arctic High-Speed Diesel (HSD) Pour point: -50°C (IS 1460 Annexure F arctic grade) · Required for all station heating & DGs	<i>NCPOR Fuel Procurement Specification</i>
Aviation Fuel (Jet A-1 / ATF) Anti-icing additive (DiEGME) treated · Essential for Basler BT-67 ski-planes & Kamov Ka-32 helicopters	<i>DGCA / DROMLAN Polar Flight Manual</i>
Emergency Freeze-Dried Rations Caloric requirement: 3,800 kcal/man-day · High-fat / high-protein retort packs with 5-year shelf-life	<i>Defense Food Research Laboratory (DFRL Mysore)</i>
Medical Oxygen & Blood Plasma 150-bar pressurized medical gas cylinders with frost-protected regulators for emergency surgery bay	<i>AFMS Polar Medical Handbook</i>
Replacement 125 kVA Alternator Heated winding insulation against condensation · Dual-redundancy baseline for life-support survival	<i>CPWD Electrical Antarctic Specifications</i>
Extreme Cold Clothing (ECWC) 4-layer wind-proof Gore-Tex & down systems rated for -60°C ambient temperatures with Sorel boots	<i>DRDO / DIPAS High-Altitude Standards</i>

Emergency Response, SAR & Safety Protocols

Emergency workflows in the Polar Command Center implement international Search and Rescue (SAR) standards established under the Council of Managers of National Antarctic Programs (COMNAP) and Antarctic Treaty.

KATABATIC BLIZZARD SHELTER-IN-PLACE Trigger: Sustained winds exceeding 45 knots or surface visibility under 200 meters Response Action: Station initiates red alert lockdown. Outdoor movements prohibited without safety tether line and buddy-system. All personnel must report to designated quarters within 15 minutes. Automatic life-line check-in timer activates. <i>Governing Framework: COMNAP Safety Manual Section 4.2 / IMD Blizzard Protocol</i>	COMNAP PROTOCOL
CREVASSE VEHICLE EXTRACTION & RESCUE Trigger: PistonBully track breakthrough or snowmobiler bridge collapse along continental ice shelf Response Action: Deployment of ITBP-led search team with ice-penetrating radar, 200m static climbing ropes, and mechanical pulley haul systems. Air ambulance Basler BT-67 alerted at Novo Airbase for standby medevac. <i>Governing Framework: Antarctic Traverse Safety Handbook & ITBP SAR Directive</i>	COMNAP PROTOCOL
STATION POWER HOUSE LOSS / DG FAILURE Trigger: Generator trip with sub-zero freeze threat to building heating circuits and Priyadarshini water line Response Action: Automatic failover to secondary 125 kVA Caterpillar genset. If secondary fails, emergency boiler runs on reserve gravity fuel tank. Station life-support circuits isolated to conserve thermal envelope. <i>Governing Framework: NCPOR Station Engineering Winter Contingency SOP</i>	COMNAP PROTOCOL
AEROMEDICAL EVACUATION (MEDEVAC) Trigger: Life-threatening trauma, acute abdominal emergency, or severe hypothermia exceeding station medical bay Response Action: AFMC Medical Officer initiates international SAR mutual assistance via COMNAP Secretariat. Basler BT-67 or Twin Otter chartered from Novo Runway / Princess Elisabeth Station to transfer patient to Cape Town hospital. <i>Governing Framework: COMNAP Medical Evacuation Guidelines & DROMLAN SAR Network</i>	COMNAP PROTOCOL

Fail-Safe Armed SOS Distress Workflow in Polar Command Center

The Polar Command Center features an Armed SOS Distress Trigger on the global top bar. In compliance with real-world polar safety:

1. Two-Step Safety Gate: Accidental triggers are eliminated via a confirm-to-arm physical slider.
2. Immediate Multi-Channel Broadcast: Dispatches priority emergency packet to all active console stations and sounds an audible alert.
3. Automatic Incident Creation: Generates an immutable emergency record with timestamp, operator ID, and GPS coordinates.
4. Resilient Offline Queuing: Operates with simulated satellite/HF blackout resilience with offline queuing and batch sync.

Authoritative URL Repository & Operational Data Integrity

Every data point in the application is auditable. Below is the complete URL verification table followed by structured answers to common technical and data architecture questions.

ORGANIZATION / PORTAL NAME	OFFICIAL URL / LINK	VERIFICATION PURPOSE IN PROJECT
NCPOR Meteorological Portal	https://www.data.ncpor.res.in/	Maitri & Bharati weather dataset listings
National Polar Data Centre	https://npdc.ncpor.res.in/pdc/	National polar scientific & geospatial catalog
NCPOR AWS Data Archive	https://npdc.ncpor.res.in/pdc/Aws/iig/Awsdata-iig.jsp	Maitri & Bharati 2012–2024 AWS benchmark records
NCPOR Research Stations	https://npdc.ncpor.res.in/npdc/research-stations.action	Official WGS-84 coordinates, elevations, capacities
Indian Antarctic Expeditions	https://ncps.ncpor.res.in/expedition/india_antarctica.php	ISEA expedition history, charters, timelines
NCPOR Polar Advisory	https://ncaor.gov.in/pages/display/352-advisory	Medical standards, clothing, rations, logistics governance
Open-Meteo Archive API	https://open-meteo.com/en/docs/historical-weather-api	ECMWF ERA5-Land historical hourly reanalysis
COMNAP Secretariat	https://www.comnap.aq/	Antarctic station catalog and international SAR manuals
Official GitHub Repository	https://github.com/MdSahilCseABES/Polar_Command_Center	Full source code, production build & CI/CD workflows

Technical Architecture FAQ: Mission Data Integrity & Telemetry Standards

Q: How is data integrity guaranteed across polar operations?

All station coordinates, elevations, capacities, blizzard thresholds, and historical weather baselines are 100% verified from official NCPOR, MoES, and IMD records. Historical weather uses ECMWF ERA5-Land reanalysis. Personnel manifests and emergency incident drills represent operational compliance models to protect personnel privacy under MoES guidelines.

Q: How does the system handle real-time GPS vs field asset telemetry?

Station coordinates (Maitri, Bharati, Himadri) are official WGS-84 fixed geodetic references. Moving field assets utilize simulated VHF telematic beacons coupled with persistent offline queuing to survive polar satellite blackouts.

Q: Why utilize numerical weather prediction rather than unverified sensor streams?

Active continuous satellite data links from Indian Antarctic stations are proprietary MoES intranets subject to severe ionospheric polar blackouts. Rather than fabricating unverified sensor feeds, the system integrates verified Open-Meteo NWP models and authentic NCPOR AWS benchmark archives for robust failover.